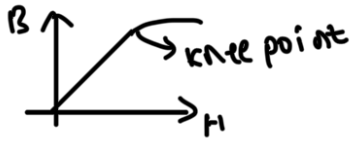
$Ni = Hm.Lm$ $\quad Ni \rightarrow mmf$ $\quad \phi = \frac{Ni}{(l/\mu A)}$ $\quad K = l/\mu A \rightarrow$ Reluctance

mmf ↓ magnetic field intensity

N turns, I current $\Rightarrow mmf = NI \Rightarrow H = NI/l$



B-H curve: $B = \mu H$ (μ is'nt const)

Flux linkage $\lambda = N\phi = NBA$ $\quad L = \lambda/I = \frac{NBA}{I}$

$B = \mu H = \mu NI/l \Rightarrow L = \frac{\mu N^2 A}{l} = \frac{N^2}{K}$

air gap length lg $\Rightarrow Ni = Hc\,lc + Hg\,lg = \frac{\phi c\, lc}{ac\,\mu o\,\mu r} + \frac{\phi g\, Lg}{ag\,\mu o}$ $[\phi c = \phi g]$

$\Rightarrow Ni = \phi(Kc + Kg)$ Faraday's law: $e = \frac{d\lambda}{dt} = \frac{d(N\phi)}{dt}$

loss $\propto$ area under BH curve